

Irreducibility of an arithmetic-progression polynomial

Theorem 1. *For every integer $n \geq 2$, the polynomial*

$$f_n(x) = \sum_{k=1}^n kx^{n-k} = x^{n-1} + 2x^{n-2} + \cdots + (n-1)x + n$$

is irreducible in $\mathbb{Q}[x]$ and in $\mathbb{Z}[x]$.

The proof compares a local estimate for the resultant of two putative factors with an estimate coming from the narrow annulus containing their complex roots. All logarithms are natural, and $\text{rad}(m)$ denotes the product of the distinct prime divisors of m .

1 Roots and the arithmetic of a factor

Put $N = n + 1$. Direct multiplication gives

$$(x-1)f_n(x) = \sum_{j=1}^n x^j - n, \quad F(x) := (x-1)^2 f_n(x) = x^N - Nx + n. \quad (1)$$

Since $f_n(1) = nN/2$, the root 1 of F has multiplicity exactly two. A common root of F and F' must be 1, so f_n has simple roots.

Lemma 2. *Every root α of f_n satisfies*

$$1 < |\alpha| < (2n+1)^{1/n}.$$

Moreover, $|\alpha|^n > 4$ if $n \geq 3$, and $n \log |\alpha| > 11/6$ if $n \geq 8$.

Proof. If $|\alpha| \leq 1$, the first identity in (1) gives $n \leq \sum_{j=1}^n |\alpha|^j \leq n$; equality forces $\alpha = 1$. The upper bound follows from $\alpha^n = N - n/\alpha$.

There are no positive real roots. For an upper-half-plane root $\alpha = \rho e^{i\theta}$, the last identity gives $n\theta = 2\pi k + \phi$ with $0 < \phi < \pi/2$. If $k = 0$, every nonconstant term of $f_n(\alpha)$ has positive imaginary part, a contradiction. Thus $\theta > 2\pi/n$. When $\theta < \pi/2$,

$$\rho^n \geq N \sin \theta > 4N/n > 4;$$

otherwise $\rho^n > N$. Negative real roots also satisfy $\rho^n > N$, and conjugation handles the lower half-plane. For $n \geq 8$, the acute-angle estimate improves to $\rho^n > N \sin(2\pi/n) > 2\pi > e^{11/6}$. Here $\sin t \geq t - t^3/6$ and $n^2/(n+1) \geq 64/9 > 2\pi^2/3$ prove the middle inequality. $\square$

Suppose henceforth that $f_n = gh$, where $g, h \in \mathbb{Z}[x]$ are monic and nonconstant. Write

$$d = \deg g, \quad a = g(0), \quad b = h(0), \quad g_j = [x^j]g, \quad J = g_1 - ag_{d-1}.$$

Conjugate root pairs and negative real roots show that $a, b, g(1), h(1)$ are positive integers. Lemma 2 gives $a, b > 1$.

Lemma 3. *For every such factorization,*

$$ab = n, \quad (a, b) = (a, g_1) = (a, J) = 1, \quad (2)$$

$$a^N > (\sqrt{nN})^d, \quad a^N \equiv (-1)^d \pmod{N}, \quad a^n \equiv 1 \pmod{b}, \quad (3)$$

$$0 < |J| \leq K_n a \log a, \quad K_n = \frac{2 \sinh(L/n)}{L/n}, \quad L = \log(2n+1), \quad (4)$$

$$|\text{Disc}(g)| |\text{Res}(g, h)| a g(1) = (nN)^d. \quad (5)$$

Here $K_n < 21/10$ for $n \geq 6$, and $K_n < 201/100$ for $n \geq 1000$.

Proof. The linear coefficient gives $ah_1 + bg_1 = n - 1$, proving (2). In particular $n \geq 6$. There are no real roots when n is odd, by (1); hence nd is always even. With $r = n/N$, multiplication over the roots of g gives $a^N = N^d g(r)$. The inequality

$$|z - r|^2 - r|z - 1|^2 = (1 - r)(|z|^2 - r) > 0 \quad (|z| > 1)$$

and $g(1) \geq 1$ give the strict norm bound. Reducing the same identity modulo N gives the first congruence. In $(\mathbb{Z}/b\mathbb{Z})[x]/(g)$, the element x is a unit and $x^n = 1$; its determinant is $(-1)^d a$, giving the second congruence.

The root formula $J = a \sum_{\alpha} (\alpha - \alpha^{-1})$, paired with its complex conjugate, yields $|J| \leq a \sum_{\alpha} 2 \sinh(\log |\alpha|) \leq K_n a \log a$. The stated constants follow from $\sinh t/t \leq (1 - t^2/6)^{-1}$ for the relevant $0 < t < 1/2$. Finally, $f'_n(\alpha) = nN/[\alpha(\alpha - 1)]$ at every root of f_n . Multiplication over the roots of g proves (5). $\square$

2 The local resultant estimate

For a prime $p \mid nN$, let M be the member of $\{n, N\}$ divisible by p , and write $M = p^r m$ with $p \nmid m$. Work in a finite unramified extension $E/\mathbb{Q}_p$ containing the m th roots of unity, with valuation $v(p) = 1$ and value group $\mathbb{Z}$. Such a field is obtained by taking a suitable finite residue field and lifting its roots of unity by Hensel's lemma.

Lemma 4. *Define $C_p(0) = C_p(1) = 0$ and $C_p(r) = p^{r-1} + C_p(r-2)$ for $r \geq 2$. Then*

$$v_p \text{Res}(g, h) \leq m C_p(r) - 2 \lfloor r/2 \rfloor. \quad (6)$$

In addition,

$$\text{rad}(bN) \mid J, \quad d \equiv \mathbf{1}_{2 \mid a} \pmod{2}. \quad (7)$$

Proof. The reductions of F are $(x^m - 1)^{p^r}$ if $p \mid N$, and $x(x^m - 1)^{p^r}$ if $p \mid n$. In the latter case there is one residue-zero root, of valuation r ; it belongs to g precisely when $p \mid a$.

Here is the Newton polygon calculation for the unit roots. The binomial valuations are

$$\begin{aligned} v_p \binom{N}{j} &= r - v_p(j), & p \mid N, \quad 1 \leq j \leq p^r, \\ v_p \binom{n+1}{j} &= r - v_p(j(j-1)), & p \mid n, \quad 2 \leq j \leq p^r + 1. \end{aligned}$$

They follow from $\binom{M-1}{k} \equiv (-1)^k \pmod{p}$ for $k < p^r$. At a nontrivial m th root of unity ζ , the negative-slope vertices of $F(\zeta + y)$ are $(0, r), (p, r-1), \dots, (p^r, 0)$. For odd p these give irreducible blocks of degrees

$$e_1 = p, \quad e_j = p^{j-1}(p-1) \quad (2 \leq j \leq r),$$

whose roots have displacement valuation $1/e_j$ from ζ . At $\zeta = 1$, remove the double root using $f_n(1+y) = \sum_{k=0}^{n-1} \binom{n+1}{k+2} y^k$; the first degree becomes $p-2$, with all later degrees unchanged. The Newton polygon theorem gives these factors in $E[x]$. They are irreducible because a factor with s roots of valuation $1/e$ has constant-term valuation $s/e \in \mathbb{Z}$.

For $p = 2$, a nontrivial residue class has one irreducible quadratic when $r = 1$. When $r \geq 2$, its first two edges merge into a quartic of displacement valuation $1/2$, followed by irreducible blocks of degrees $4, 8, \dots, 2^{r-1}$. The quartic's factors have even degree by the slope denominator two. At residue one the degrees are $2, 4, \dots, 2^{r-1}$ for $r \geq 2$, with no block for $r = 1$. This lists all $n - 1$ roots.

First consider odd p . A rational factor selects whole blocks. Different residue classes contribute zero to the resultant valuation; two blocks $i < j$ in the same class contribute e_i if assigned to different factors. Thus we must bound a graph cut with edge weight e_i between $i < j$. Its maximum is $C_p(r)$: the last two vertices can be placed in opposite parts since $e_{r-1} \geq \sum_{i < r-1} e_i$; deleting them contributes exactly $\sum_{i < r} e_i = p^{r-1}$. At residue one this sum is $p^{r-1} - 2$, so the bound is $C_p(r) - 2\lfloor r/2 \rfloor$.

For the binary quartic P , the identity $F'(\alpha) = nN(\alpha - 1)/\alpha$ gives $v(F'(\alpha)) = r$, and the $r - 2$ higher blocks each contribute one to this valuation. Thus $v(P'(\alpha)) = 2$ and $v(\text{Disc } P) = 8$. If P splits between g and h , both parts are quadratic. After translation, each has constant coefficient of valuation one and linear coefficient of valuation at least one; its discriminant has valuation at least two. The resultant between the two parts therefore has valuation at most $(8 - 2 - 2)/2 = 2$. Two formal vertices of weight two, joined by an edge of weight two, bound every assignment of this quartic. The same cut calculation now applies to the sequence $2, 2, 4, 8, \dots$; at residue one, insert an initial vertex of weight zero. Summing over the m residue classes proves (6).

For $p \mid bN$, all roots of g are units. Reduce in the residue field of a valued splitting extension. Every selected block at a nontrivial residue root has degree divisible by p , so its sum of $\alpha - \alpha^{-1}$ reduces to zero; roots at residue one also contribute zero. Thus $v_p(J) > 0$, and integrality gives $v_p(J) \geq 1$. At $p = 2$ all unit blocks have even degree, proving the parity assertion as well. $\square$

Put $R = \log |\text{Res}(g, h)|$, $B = \frac{1}{2} \log(nN)$, and $Q = \prod_{p^2 \mid nN} p$. Formula (5) shows that the resultant has no prime divisor outside nN . Lemma 4 gives

$$R \leq \left(\frac{N}{8} - 2 \right) B + NC_8 \quad (n \geq 15), \quad R \leq N \sum_{p \mid Q} \frac{p \log p}{p^2 - 1}, \quad (8)$$

where

$$C_8 = \frac{3}{8} \log 2 + \frac{5}{24} \log 3 + \frac{3}{40} \log 5 + \frac{1}{56} \log 7 < \frac{13}{20}.$$

Indeed, $C_p(r)/p^r = \sum_{j=1}^{\lfloor r/2 \rfloor} p^{-(2j-1)}$. Subtract $1/8$ from each summand; positive excess occurs only in the first layers for $p = 2, 3, 5, 7$. With $B_0 = \sum_p \lfloor v_p(nN)/2 \rfloor \log p \leq B$, this gives $R \leq (N/8 - 2)B_0 + NC_8$. The second bound follows from $C_p(r) < p^{r+1}/(p^2 - 1)$, and $C_p(1) = 0$.

3 Excluding small constant terms

Lemma 5. *If $n \geq 2$ and $n + 1$ is a prime power, then f_n is irreducible.*

Proof. Suppose $N = p^r$ and choose g of smaller degree, so $0 < d < N/2 \leq \varphi(N)$. Modulo p , $f_n = (x - 1)^{N-2}$. Every root α of g has $v_p(\alpha - 1) \geq 1/d > 1/\varphi(N)$, since its ramification index is at most its local field degree, which is at most d . Eisenstein's criterion for $\Phi_N(x + 1)$ gives $v_p(\zeta - 1) = 1/\varphi(N)$ for primitive N th roots ζ . Hence $v_p(\Phi_N(\alpha)) = 1$. Also $f_n(\zeta) = N/(1 - \zeta)$, so

$$|\text{Res}(f_n, \Phi_N)| = N^{\varphi(N)}/p, \quad |\text{Res}(g, \Phi_N)| = p^d.$$

The second equality follows because it divides the first integer and has p -adic valuation d .

Let $H = \sqrt{nN}$. For odd p and $r \geq 3$, Lemma 2 gives

$$|\Phi_N(\alpha)| \geq \frac{|\alpha|^N - 1}{|\alpha|^{N/p} + 1} > \frac{3}{8} |\alpha|^{\varphi(N)}.$$

Multiplying and using (3) yields $p > \frac{3}{8}H^{1-1/p}$, whereas $H > p^3 - 1 > \frac{8}{9}p^3$ implies $H^{1-1/p} > \frac{8}{9}p^2 \geq \frac{8}{3}p$. For $p = 2$, $\Phi_N(x) = x^{N/2} + 1$ gives instead $2^d > a^{N/2}/2^d > (\sqrt{H}/2)^d$, hence $H < 16$. This excludes $N \geq 32$. For $N = 4, 8$ the constant n is prime; for $N = 16$ the constants would be 3, 5, contradicting $5^{15} \equiv 1 \pmod{3}$ in (3).

If $N = p^2$ with odd p , one factor has even degree and the other odd degree. Their constants are respectively 1 and -1 modulo p by (3), so they are at least $p + 1$ and $p - 1$. Their product $p^2 - 1$ forces equality, contradicting coprimality. Finally, if $N = p$ is odd prime, $f_n(x + 1)$ is Eisenstein at p . The value $N = 2$ is excluded by $n \geq 2$. $\square$

Lemma 6. *Every nontrivial monic integer factorization has $a, b \geq 13$; in particular, $n \geq 182$.*

Proof. The integers a, b, N are pairwise coprime. By Lemma 5, N has at least two distinct prime divisors. Thus

$$30 \leq \text{rad}(bN) \leq |J| < \frac{21}{10}a \log a,$$

and the radical contains at least three primes coprime to a . This excludes $2 \leq a \leq 7$, since $\log 7 < 2$. For $a = 8, 9, 10, 12$, the least possible products of three such primes are 105, 70, 231, 385, respectively; $\log a < 3$ excludes each.

If $a = 11$, then $\text{rad}(bN) < 70$. This radical is even, is coprime to 11, and has at least three prime divisors, hence is 30 or 42. If n is even, these supports force b to be a power of two and $3 \mid N$. By (7), d is even, so $11^N \equiv 1 \pmod{3}$ contradicts odd N . Thus b is odd. For its least prime divisor q , the order of 11 modulo q divides 11: it divides $11b$ by (3) and divides $q - 1$, which is coprime to b . Order 11 would force $q \geq 23$, outside the possible supports; order one forces $q = 5$. Since N is not a prime power,

$$b = 5^u, \quad N = 2^v 3^w, \quad u, v, w \geq 1.$$

The equality $11 \cdot 5^u + 1 = 2^v 3^w$, reduced modulo 3, 8, 5 in turn, gives u even, $v = 2$, and $w \equiv 2 \pmod{4}$. Therefore

$$(2 \cdot 3^{w/2} - 1)(2 \cdot 3^{w/2} + 1) = 11 \cdot 5^u.$$

The left factors are coprime odd integers greater than one, so they must be 11 and 5^u . Their difference two is impossible. Interchanging g, h proves the claim. Since $(a, b) = 1$, their product is at least $13 \cdot 14 = 182$. $\square$

4 The archimedean estimate and the contradiction

Choose g of smaller degree and put

$$\ell = \log n, \quad L = \log(2n + 1), \quad A = \log a, \quad \delta = d/n.$$

Then $A \geq \log 13 > 51/20$ and $0 < A/L < \delta < 1/2$.

Lemma 7. *If the logarithms of the root moduli of g lie in an interval of width at most W/n , where $W > 0$, then*

$$G_W := \delta(\ell - A + \log W) - \frac{1}{2}W\delta^2 \leq \frac{R + \log g(1)}{n} - \delta \log(N/n). \quad (9)$$

Proof. Write $\alpha_i = \rho_i u_i$, with $|u_i| = 1$, and set $\sigma = W/n$, $t = e^{-\sigma}$. For $i \neq j$,

$$\frac{|\alpha_i - \alpha_j|^2}{\rho_i \rho_j} \leq t^{-1} |1 - t u_i \overline{u_j}|^2.$$

The absolutely convergent logarithmic series gives

$$\sum_{i \neq j} \log |1 - t u_i \overline{u_j}| = - \sum_{k \geq 1} \frac{t^k}{k} \left| \sum_i u_i^k \right|^2 - d \log(1 - t).$$

Summing the pairwise bound, and using $1 - e^{-\sigma} \geq \sigma e^{-\sigma/2}$, yields

$$\log |\text{Disc}(g)| \leq (d-1)A + \frac{1}{2}\sigma d^2 - d \log \sigma.$$

Insert this in (5) and divide by n . □

Lemma 2 permits $W = L$ and, for $n \geq 8$, $W = L - 11/6$. Since $\log g(1) \leq \log(nN/2) = 2B - \log 2$, the term $-2B$ in (8) cancels. For either width and $n \geq 182$, this gives

$$G_W < \ell/8 + 33/50. \tag{10}$$

Indeed, $B < \ell + 1/(2n)$, $C_8 < 13/20$, and the error above $\ell/8 + 13/20$ is at most $(\ell/8 + 13/20 + 1/16 + 1/(16n))/n < 1/100$.

There is a second bound when $n \geq 1000$. Put $K = 201/100$ and

$$S(A) = \frac{3}{20}A + \frac{1}{10} \log(KA) + \frac{97}{100}.$$

The primes of Q dividing a have product at most $\sqrt{a}$; all other primes of Q divide bN . Therefore $Q \leq \sqrt{a} \text{rad}(bN) < Ka^{3/2}A$. For $p \geq 11$, $p/(p^2 - 1) \leq 1/10$, while

$$\sum_{p=2,3,5,7} \left(\frac{p}{p^2 - 1} - \frac{1}{10} \right) \log p < \frac{97}{100}.$$

Consequently (8) and (9) imply

$$G_L < S(A) + 1/50. \tag{11}$$

For the error estimate, also use $Q \leq \sqrt{nN}$: the extra terms are less than $(2.1\ell + 1)/n < 1/50$.

We finish by recording the elementary comparisons against these two upper bounds; the details are included to specify the constants.

Lemma 8. *Let $n \geq 182$ and $51/20 \leq A < \delta L < L/2$. Take $W = L - 11/6$ if $n < 1000$, and $W = L$ otherwise. The bound (10), together with (11) when $n \geq 1000$, is impossible.*

Proof. First let $n \geq 1000$. Then $\ell > 6.9$, $0 < L - \ell < 0.7$ and $\log L > 2$. The expression G_L is increasing in δ on the stated interval, so

$$G_L > H(A) := \frac{A(\ell + \log L - 3A/2)}{L}.$$

The following bounds cover all possible A :

range of A	lower bound
$[51/20, 3]$	$H(A) - S(A) > 1/10$
$[3, \ell/4]$	$H(A) - S(A) > 51/200$
$[\max(3, \ell/4), L/2]$	$H(A) - \ell/8 > 59/80$.

An interval with reversed endpoints is ignored. For the first row, $(\ell + 2 - 1.5A)/(\ell + 0.7)$ increases with ℓ on this range, so

$$H(A) > H_0(A) := \frac{A(8.9 - 1.5A)}{7.6}.$$

The function $H_0 - S$ is concave on this interval; at its endpoints it exceeds 0.18 and 0.1, respectively, using $\log(2.55K) < 1.7$ and $\log(3K) < 2$. For the second row, $H(A) > 5A/8$, and $19A/40 - \log(KA)/10 - 0.97$ is increasing for $A \geq 3$ and exceeds 0.255 at three. For the last row use concavity of H . At $A = L/2$,

$$H(A) - \ell/8 = \frac{1}{2} \log L - \frac{3}{8}(L - \ell) > 59/80.$$

At the other endpoint, if $6.9 \leq \ell \leq 12$, its lower bound is $3(\ell - 2.5)/(\ell + 0.7) - \ell/8$. This is concave in ℓ ; its endpoint values are $1329/1520$ and $189/254$, both greater than $59/80$. If $\ell \geq 12$, the lower bound is $\ell(\ell/32 + 33/80)/(\ell + 0.7) \geq 189/254$. The first two rows contradict (11), and the last contradicts (10).

Now let $182 \leq n < 1000$ and take $W = L - 11/6$. Write $c = L - \ell$. Then $5.2 < \ell < 7$, $0.69 < c < 0.7$ and $1.4 < \log W < 1.8$. Again G_W increases with δ , so

$$G_W > H_W(A) := \frac{A(\ell + \log W)}{L} - \frac{A^2(L + W/2)}{L^2}.$$

For $51/20 \leq A \leq L/2$, we have $H_W(A) \geq H_W(L/2)$. To check this, factor their difference as

$$H_W(A) - H_W(L/2) = \frac{A - L/2}{L} T, \quad T = \ell + \log W - (A + L/2)(1 + W/(2L)).$$

The bracket satisfies

$$2LT \leq P(\ell, c) := \frac{1}{2}\ell^2 - \ell c - \frac{3}{2}c^2 - \frac{47}{15}(\ell + c) + \frac{187}{40} < 0.$$

Indeed, P decreases in c and $P(\ell, 0.69)$ increases in ℓ on this rectangle; its maximum is

$$P(7, 0.69) = -27869/60000 < 0.$$

It follows that

$$G_W > H_W(L/2) = \ell/8 + \frac{1}{2} \log W - \frac{3}{8}c + \frac{11}{48} > \ell/8 + \frac{2}{3},$$

contradicting (10), since $2/3 > 33/50$. All terminating decimals above are exact rational constants; the stated logarithmic bounds follow from the exponential series. $\square$

Proof of Theorem 1. By Gauss's lemma, a reducible monic f_n has a nontrivial monic integer factorization. Lemma 6 forces $n \geq 182$. Choosing its smaller-degree factor gives $51/20 \leq A < \delta L < L/2$, while the local resultant and discriminant estimates give (10) and (11) in their stated ranges. Lemma 8 is a contradiction. Thus f_n is irreducible over $\mathbb{Q}$, and its monicity gives irreducibility in $\mathbb{Z}[x]$ as well. $\square$